

Abhitej Divi

<https://github.com/abhitejdivi5/projects>

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PROFESSIONAL SUMMARY

Analog IC Designer experienced in full-cycle ASIC tapeout, including design and verification of SAR ADC, amplifiers, bandgap references, and LDO's. Skilled in schematic entry, post-layout verification, and circuit simulation using Cadence Virtuoso. Seeking opportunities in the analog IC design domain.

EDUCATION

- **Prairie View A&M University** Prairie View, TX
Doctor of Philosophy in Electrical and Computer Engineering *Aug 2022 – Aug 2025*
- **Lamar University** Beaumont, TX
Master of Engineering in Electrical Engineering *Aug 2021 – Aug 2022*
- **Lakireddy Bali Reddy College of Engineering** Mylavaram, INDIA
Bachelor of Engineering in Electronics and Communication Engineering *Aug 2016 – Jul 2020*

WORK EXPERIENCE

- **Apple New Silicon Initiative (NSI) Graduate Research Assistant** Prairie View, TX
Prairie View A&M University *Jan 2023 – Aug 2025*
 - Executed Apple-funded ASIC tapeout projects, from design specifications to final GDSII submission, ensuring schematic accuracy and design rule compliance.
 - Experienced in Cadence AMS design and verification, including RTL modeling, lint checks, co-simulation, and hierarchical mixed-signal design optimization.
 - Collaborated with cross-functional analog, layout, and verification teams for DRC/LVS validation and timing closure.
 - Served as Technical Advisor for the Computer Hardware and VLSI Design Club; led workshops, mentored students, and created lab manuals for analog and digital circuits.
- **Graduate Teaching Assistant** Beaumont, TX
VLSI CAD & Signal Integrity Laboratory, Lamar University *May 2022 – Jul 2022*
 - Worked on soft error and soft delay mitigation using a dynamic threshold technique. Assisted with Electronics-I (3421) instruction, grading.

SKILLS

- **Key Skills:** Amplifiers, Current Mirrors, Two-Stage Amplifiers, Compensation Techniques, Common-Mode Feedback, Bandgap References, Sampling Circuits, Strong-Arm Latch Comparator, ADCs, Tapeout
- **EDA Tools:** Cadence Virtuoso (Schematic & Layout), ADE Explorer, Assura (DRC/LVS/ERC), Spectre, PVT Corner Analysis, Genus, Innovus, MATLAB, LabVIEW, HSPICE, Verilog, AutoCAD, Unix/Linux
- **Courses & Certifications:**
 - High-Speed Analog Design, Mixed-Signal IC Design (SSCD Lab, IIT Kanpur – Online Lectures)
 - Analog Circuit Design and Simulation Onboarding (vIC23.1/IC6.1.8) and Cadence RTL-to-GDSII Flow Training

- Diagnostic ASIC Tapeout (Apple NSI Tape-Out)** *Mar 2025*
 - Designed a diagnostic ASIC operating at 150 MHz, integrating amplifiers, digital logic, and memory units, with layouts following symmetry and common-centroid constraints.
 - Performed post-layout verification, including DRC, LVS, and ERC checks. Applied shielding, guard rings, and seal rings to minimize noise coupling.
 - Implemented a 32-pin ESD-protected pad frame; completed routing and finalized GDSII submission for fabrication.

- Fully Differential 10-bit Asynchronous SAR ADC** *May 2024*
 - Designed a fully differential 10-bit asynchronous SAR ADC achieving 100 MHz sampling rate with monotonic switching for the capacitive DAC.
 - Achieved 10-bit ENOB and 61.8 dB SQNR from spectrum analysis of the digital output.
 - Implementing a Pipelined ADC using the designed SAR ADC (ongoing work).

- Bootstrapped Switch and Strong-Arm Latch Comparator Design** *May 2024*
 - Implemented a high-linearity bootstrapped switch for the sample-and-hold network and verified performance with large capacitive loads.
 - Verified a dynamic strong-arm latch comparator operating at 1 GHz for SAR ADC applications.

- Fully Differential Two-Stage Op-Amp with Common-Mode Feedback (CS 2nd Stage)** *Jan 2024*
 - Implemented a first-stage 5T OTA achieving 71 dB DC gain, 63° phase margin, and 236 MHz UGB.
 - Designed a telescopic first stage achieving 99 dB DC gain, 63° phase margin, and 243 MHz UGB.
 - Compared both topologies for gain, bandwidth, and power efficiency to evaluate trade-offs for high-speed applications.

- Op-Amp based Bandgap, fractional bandgap and LDO Design** *May 2023*
 - Designed a self-biased, opamp-based bandgap reference with an integrated startup mechanism generating a constant 1.25V.
 - Designed Fractional Bandgap generating constant 0.6V
 - Designed LDO generating constant 1.6V. All designs verified for temperature -45°C to 125°C and supply variations.

PUBLICATIONS AND CONFERENCES

- Solar Photovoltaic Diagnostic System with Logic Verification and Integrated Circuit Design for Fabrication, *Solar*, Vol. 5, No. 2, MDPI, 2025. Divi Abhitej, and Shuza Binzaid. (Utility Patent Submitted)
- Simplifying Complex Digital Sequential Circuits by an Innovative Mixed-Signal Circuit Alternative, *Computers & Electrical Engineering*, Elsevier, 2024. Shuza Binzaid, Abhitej Divi, Md. Rokonzaman.
- Presented Apple-funded tape-out work at the CHIPS Semiconductor Summit (College Station, TX) and the NSI@HBCUs Summer 2024 Convening (Apple Park, Cupertino, CA).
- A 120-MS/s Fully Differential 10-Bit SAR ADC Employing a Low-Power Monotonic Switching DAC Architecture. (Submitted - IEEE Transactions)